

BST Physics Curriculum Vol 11 Chapter 4 — Holomorphic Discrete Series + Highest Weights v0.4 SIGNATURE (refilled per Cal #104)

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Vol 11 Chapter 4 — Holomorphic Discrete Series + Highest Weights

Chapter motivation

Standard representation theory of semisimple Lie groups: irreducible unitary representations classified into discrete series (square-integrable matrix coefficients) + principal series (induced from parabolic subgroups) + supplementary series. Harish-Chandra 1955-1976 + Knapp 1986 develop the theory. Discrete series exist only when G has compact Cartan subgroup; classified by Harish-Chandra parameters; for hermitian symmetric pairs (G, K) with G/K bounded HSD, the holomorphic discrete series are realized in Bergman spaces $H^2(D)$ on the bounded domain.

BST substrate-cross-link: per Vol 11 Ch 3 + Wallach 1976, Bergman $H^2(D_{IV}^5)$ IS holomorphic discrete series of $SO_0(5, 2)$. K-type $V_{(p,q)}$ labeled by integer pair (p, q) admissible per Wallach + Faraut-Koranyi; (p, q) corresponds to Harish-Chandra parameter labels in dual classification.

Section 4.0b — Reader-grade 3-level pedagogy

Level 1: Standard holomorphic discrete series classification (Harish-Chandra + Knapp + Vogan); BST cross-link: Bergman $H^2(D_{IV}^5)$ IS holomorphic discrete series of $SO_0(5, 2)$; Wallach K-type $V_{(p,q)}$ corresponds to highest-weight labels.

Level 2 (graduate-mathematician): Standard Harish-Chandra-Knapp theory: irreducible unitary representations of semisimple Lie group G classified into discrete series (square-integrable; exist only when G has compact Cartan subgroup) + principal series + supplementary series. Discrete series classified by Harish-Chandra

parameter λ in (regular dominant integral) coset of compact Cartan + Weyl group. For hermitian symmetric pair (G, K) with G/K bounded HSD: holomorphic discrete series realized in Bergman $H^2(D)$ on bounded domain $D = G/K$ via reproducing kernel structure (Vol 11 Ch 2). Highest-weight classification: each holomorphic discrete series has highest-weight vector $\lambda \in$ (positive Weyl chamber); representation generated by K -orbit of λ . For $SO_0(5, 2)$ type IV: holomorphic discrete series parametrized by $(n+1)$ -parameter family of highest weights; Wallach 1976 analyticity bound determines admissible parameter range. BST D_{IV}^5 : K -types $V_{(p,q)}$ under $K = SO(5) \times SO(2)$ labeled by integer pair (p, q) ; corresponds to Harish-Chandra highest weight $(p \cdot \omega_1 + q \cdot \omega_2)$ where ω_1, ω_2 are fundamental weights for K -Cartan. Ground state $V_{(0,0)} =$ scalar K -type representation; Casimir $Cas(0, 0) = 6 = C_2$ (T2439 RIGOROUSLY CLOSED). Higher-weight K -types $V_{(p,q)}$ with $p, q > 0$ have Casimir eigenvalues increasing per K -type Casimir formula. Cross-link to Vogan 1981 unitary representations theory + Knapp-Vogan 1995 cohomological induction for non-degenerate cases. Substrate-physics: each K -type $V_{(p,q)}$ is a substrate “energy level” with Casimir = energy eigenvalue (Vol 5 Ch 4 substrate Schrödinger $H_{\text{sub}}|\psi\rangle = E|\psi\rangle$); BST observables = substrate K -type expansion coefficients.

Level 3 (5th-grader accessible): Lie group representation theory classifies how groups act on vector spaces. “Discrete series” representations are special — they live in honest function spaces (like Bergman $H^2(D_{IV}^5)$). The substrate Bergman space IS the holomorphic discrete series of $SO_0(5,2)$. Wallach K -types $V_{(p,q)}$ (Vol 11 Ch 3) are the “natural building blocks” — substrate energy levels with Casimir eigenvalues.

Section 4.1 — Standard Harish-Chandra Discrete Series

Semisimple Lie group G discrete series: irreducible unitary representations with square-integrable matrix coefficients; exist only when G has compact Cartan subgroup.

Harish-Chandra parameter λ in regular dominant integral coset of compact Cartan + Weyl group.

Standard refs: Harish-Chandra 1955-1976 collected papers; Knapp 1986 *Representation Theory of Semisimple Groups*; Vogan 1981 *Representations of Real Reductive Lie Groups*.

Section 4.2 — Holomorphic Discrete Series on Bounded HSDs

For hermitian symmetric pair (G, K) with G/K bounded HSD $D = G/K$: holomorphic discrete series realized in Bergman $H^2(D)$ via reproducing kernel (Vol 11 Ch 2).

Highest-weight λ classification: each holomorphic discrete series has highest-weight vector; representation generated by K -orbit of λ .

Section 4.3 — $SO_0(5, 2)$ Holomorphic Discrete Series for D_{IV}^5

Wallach 1976: $(n+1)$ -parameter family of highest weights for type IV $SO_0(n, 2)$; analyticity bound determines admissible range.

BST D_{IV}^5 ($n = 5$): K-type $V_{-}(p, q)$ under $K = SO(5) \times SO(2)$ labeled by integer pair (p, q) ; corresponds to Harish-Chandra highest weight $(p \cdot \omega_{-1} + q \cdot \omega_{-2})$ with ω_{-1}, ω_{-2} fundamental weights for K-Cartan.

Section 4.4 — Ground State + Higher K-Types

$V_{-}(0,0)$ ground state = scalar K-type; Casimir $\text{Cas}(0, 0) = 6 = C_{-2}$ (T2439 RIGOROUSLY CLOSED Thursday).

Higher $V_{-}(p, q)$ with $p, q > 0$: Casimir eigenvalues increasing per K-type Casimir formula.

Section 4.5 — Substrate-Physics Energy Levels

Each K-type $V_{-}(p, q)$ is a substrate “energy level” with Casimir = energy eigenvalue.

Vol 5 Ch 4 substrate Schrödinger $H_{\text{sub}}|\psi\rangle = E|\psi\rangle$ with E = Casimir eigenvalues from substrate K-type spectrum.

BST observables = substrate K-type expansion coefficients.

Section 4.6 — Honest scope + Connection

- Standard holomorphic discrete series + highest-weight theory [?]
- Wallach K-type $\leftrightarrow$ Harish-Chandra highest-weight correspondence [?]
- Ground-state $C_{-2} = 6$ (T2439 RIGOROUSLY CLOSED) + higher K-type spectrum
- Open scope: explicit Casimir eigenvalue formula derivation for $V_{-}(p, q)$ (multi-week)

Connection: - Vol 11 Ch 3 Wallach K-Type Representation Theory - Vol 1 Ch 5 Casimir Algebra (substrate spectrum) - Vol 5 Ch 4 Schrödinger Equation (energy eigenvalues) - Vol 10 Ch 6 Group Theory + Lie Groups

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